

Kuntay Yilmaz

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EDUCATION

Hacettepe University

Turkey

Bachelor of Science in Artificial Intelligence Engineering

Expected Graduation: June 2027

- **GPA: 3.55**
- **Relevant Coursework:** Computer Vision, Computational Photography, Deep Learning, Machine Learning, Data Science, Differential Equations

EXPERIENCE

Undergraduate Researcher

March 2026 – Present

Hacettepe University

- Working on **memory efficient and real-time Continuous Sign Language Translation** under the supervision of Assoc. Prof. Dr. Hacer Yalim Keles.
- Investigating **model compression, knowledge distillation, and lightweight temporal modeling** to reduce GPU memory usage and latency while preserving translation quality.

Research Intern

July 2025 – August 2025

METU NLP Research Group

- Researched state-of-the-art **polygon-based building extraction** models for **aerial/satellite imagery** under the supervision of Asst. Prof. Dr. Cagri Toraman.
- Integrated the top performing model into a **Django-based annotation platform**, enabling model predicted building polygons to reduce manual labeling effort.

PROJECTS

HDR Novel View Synthesis from Multi-Exposure Multi-View Images [GitHub]

Spring 2026

- Investigating **feed-forward HDR novel view synthesis** from **multi-exposure multi-view LDR images** using a pretrained LagerNVS backbone.
- Developing an **HDR-aware adaptation strategy** to preserve LagerNVS geometry/view synthesis capabilities while improving exposure handling and HDR appearance reconstruction.

XAI Evaluation of CameraHMR for 3D Human Pose Estimation

Spring 2026

- Analyzed **CameraHMR** on **3DPW** and **EMDB**, comparing gradient, perturbation and attention-based XAI methods for SMPL pose and camera output attribution.

Neural Physics Engine (NPE) [GitHub]

November 2025

Technologies: PyTorch, Python, Pymunk

- Re-implemented the ICLR 2017 paper *A Compositional Object-Based Approach to Learning Physical Dynamics* and trained classic and modernized variants on Pymunk-generated object interaction simulations.

ResNet-50 Implementation for CIFAR-10 [GitHub]

September 2025

Technologies: PyTorch, Python

- Implemented **ResNet-50** from scratch and built a reproducible CIFAR-10 training pipeline, achieving **94.8%** test accuracy.

2D Game Engine with a Space Shooter Prototype [GitHub]

June 2024

Technologies: C++, SFML, Box2D, CMake

- Built a custom **2D game engine** using **C++** and **SFML**, integrated **Box2D** physics for collisions, and demonstrated the engine through a space-shooter prototype.

ACHIEVEMENTS & ACTIVITIES

Algorithm Competition Winter & Summer Camp 2025

2025

- Ranked **1st** in a team contest among **23** students at a one-week algorithm camp organized by **Inzva**.

TECHNICAL SKILLS

Languages & Tools: Python, C, C++, OpenGL, Git

Frameworks & Libraries: PyTorch, OpenCV, NumPy, pandas, Matplotlib

Research Interests: 3D Vision, Generative Models, Deep Learning, Real-Time Graphics